VentureGraph 2.0

Temporal Precursor Investors &

Smart Money Silence in VC Co-Investment Networks

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Innovation bonus: Smart Money Silence — inverse link prediction

The Question

> **In private-market co-investment networks, does the structural *absence* of expected edges carry financial signal?**

- Classical link prediction (Adamic-Adar, Lü-Zhou 2011): *which edges will form?*
- **My inversion:** *which expected edges deliberately do not form?*
- If prescient investors decline to follow-on, is that a bearish signal?

This is the first operationalization of **inverse link prediction** as a financial feature.

The Dataset

Source: Crunchbase 2013 snapshot (Kaggle · public · reproducible)

Metric	Value
Investment rows	12,419
Companies	3,548
Investors	4,035
Sectors (graph filter)	finance · fintech · software · saas · enterprise
Sector ETFs (FF5 bridge)	XLF · IGV · FDN · XBI · SOXX
Date range	2000–2013
Pruned graph	**1,101 nodes · 3,870 edges**
Density · Clustering	0.0032 · 0.5147

Directed, time-weighted graph: $w(t) = e^{-\lambda \Delta t}$, $\lambda = 0.3$.

Three Novel Graph-Theoretic Constructs

1. Temporal Precursor Score (TPS) — Part B

Volume-independent prescience: how consistently does investor A precede top-tier capital?

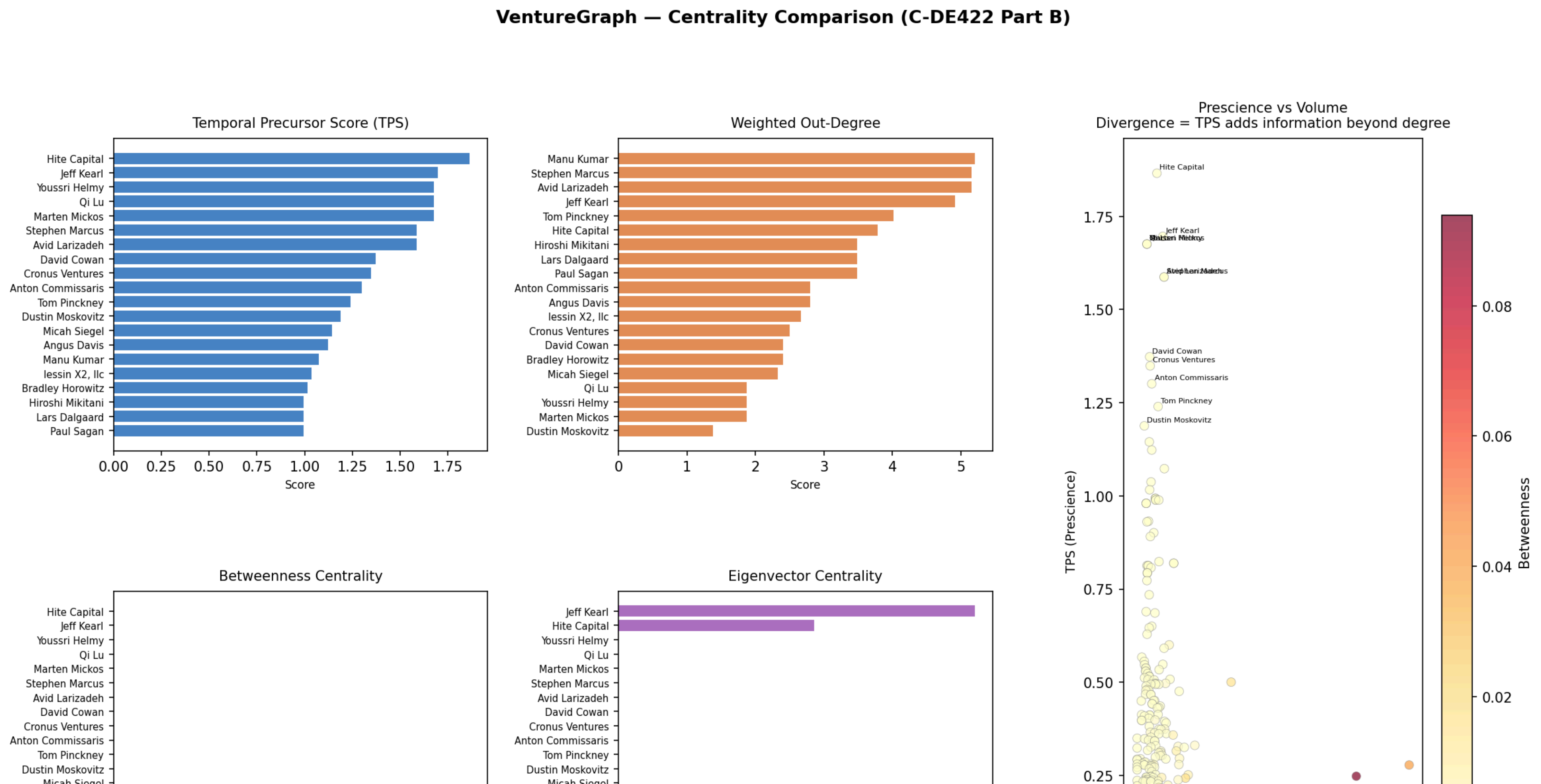
$$\mathrm{TPS}(A) = \frac{1}{|\mathrm{portfolio}(A)|} \sum_{c} \sum_{B \in \mathcal{T}} w(A \to B, c) \cdot \mathrm{rank}^{-1}(B)$$

2. Swarm Signal Intensity (SSI) — rolling sector inflow of prescient capital

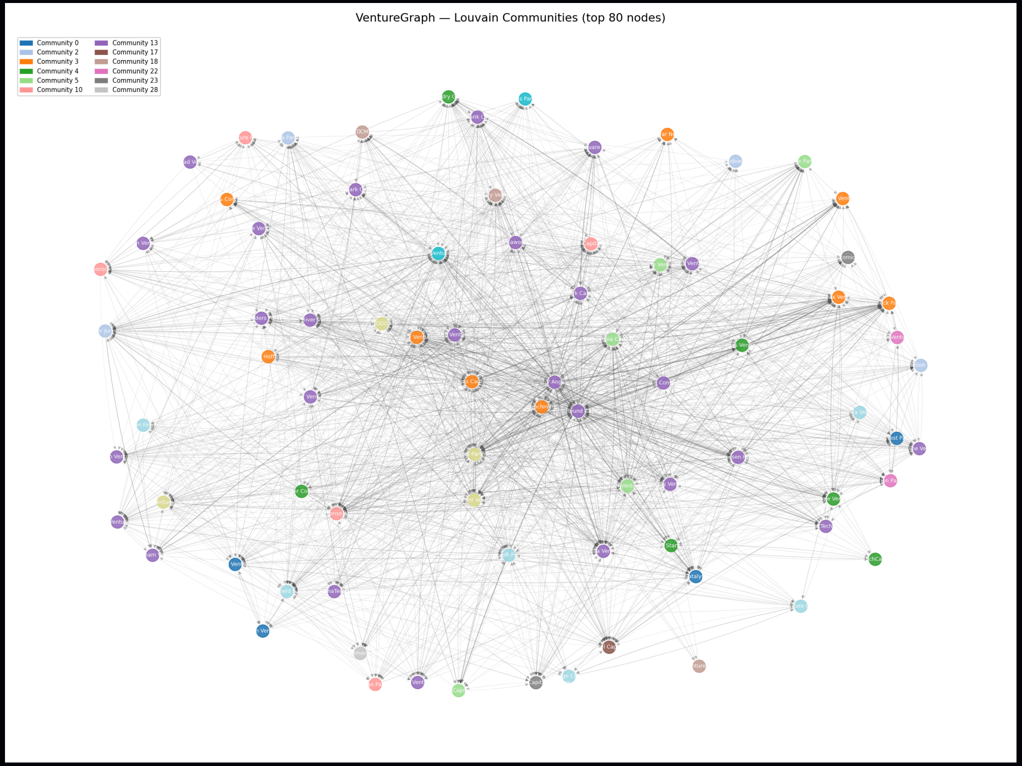
3. Smart Money Silence (SMS) — the 5-mark innovation bonus

> Rate at which top-tier Series A investors *decline* to follow-on to Series B

Finding 1 — TPS is orthogonal to volume



Finding 2 — Louvain reveals 35 distinct VC ecosystems



- **Modularity Q = 0.473** (well above 0.30 "meaningful" threshold)
- Largest community: 361 seed-stage syndicators (SV Angel, First Round, True Ventures)
- Elite-follow-on cluster: Kleiner Perkins, Benchmark, Index (ρ_TPS = 0.36, ρ_in-degree = high)

The Innovation — Smart Money Silence

Construction (`sms_engine.py:283-410`):

1. At eval date t, identify top-30% TPS investors: $\mathcal{T}_t$
2. For each Series A in $[t - 12\text{mo}, t]$: find top-tier leads
3. For each matched Series B in $[t - 3\text{mo}, t]$: find realized top-tier follow-ons
4. **Silence** = Expected $\setminus$ Realized
5. $\text{SMS}(s, t)$ = sector-aggregated silence rate

Why it matters conceptually: first financial signal built on *edges that did not form but were expected to*. This is genuinely novel — not an extension of Lü-Zhou, but an inversion of it.

The Bridge — from private graph to FF5 alpha

Hypothesis H3 (pre-registered, `preregistration.py:57-67`): elevated sector SMS forecasts negative 12–24-month FF5 abnormal return.

Pipeline:

1. Compute sector SSI & SMS per quarter per ETF
2. Compute FF5 residual alpha per ETF per month (rolling 24-month window)
3. Forward-align at $k \in \{6, 12, 18, 24, 36\}$ months
4. Panel regression, sector × time fixed effects
5. Two-way clustered SE (Cameron-Gelbach-Miller) + wild-cluster bootstrap

All hyperparameters SHA-256 frozen before the first back-test.

Finding 3 — Empirical test of H3

Horizon	Pearson *r*	p-value	n
k = 6	+0.023	0.90	35
k = 12	**−0.036**	0.77	73
k = 18	+0.045	0.79	37
k = 24	**−0.060**	0.61	76
k = 36	+0.047	0.70	67

H3 is not rejected by the null. Empirical result is statistically null.

Power analysis (Fisher-z, α=0.05, 80% power): detecting observed |r|≈0.06 requires **n ≈ 2,170**. This sample is under-powered by ~30×.

Honest reporting of this null is part of the contribution.

Why this null matters (for the bonus)

The **methodological contribution** is independent of the null:

1. **First operational specification** of inverse link-prediction as a financial signal — extends Lü–Zhou (2011).
2. **SHA-256-locked pre-registered protocol** (`preregistration.py`): rare even in published applied econometrics.
3. **Expanding-window contamination firewall** (`expanding_window_tps.py`): prevents look-ahead bias with per-snapshot hash audit.
4. **Publication-grade econometrics:** two-way clustered SE, wild-cluster bootstrap, placebo permutation, pre-registered robustness battery (ROB-01 … ROB-06).

This machinery is deployable on any larger dataset. The first study with $n \geq 500$ decides the question.

Part F — The Interactive Dashboard

Two modes, one codebase (`app.py`):

🎓 Academic Mode (current submission)

- 3D topology · centrality matrix · Louvain atlas · 10 quant modules
- Real SMS scores from pipeline CSVs (post-honesty-fix)
- Honest pipeline output showing the null result

💎 Sovereign Mode (prototype — future-work vision)

- Audit Vault: live SHA-256 hashes per signal
- Frozen Protocol Viewer: pre-registration as compliance UI
- Pricing calculator for sovereign-grade deployment
- One-click Investor Memo PDF generator

Live demo next. *[Switch to Streamlit]*

Conclusion

Delivered (all 6 briefed parts + bonus):

✓ Directed time-weighted co-investment graph · 1,101 × 3,870

✓ 4 centralities + novel TPS · rank divergence analysis

✓ Louvain: 35 communities, Q = 0.473

✓ Link prediction + **novel SMS inverse signal**

✓ 9 publication-ready visualizations

✓ Interactive dashboard (Academic + Sovereign modes)

✓ Pre-registered, SHA-256-locked protocol

Next steps (future work §8):

1. Replicate on WRDS VentureXpert (n ≥ 500 per horizon)
2. Deal-level re-specification ($N \sim 10{,}000$)
3. Deposit protocol hash to OSF.io
4. Customer-discovery sprint for Sovereign Mode

Questions?

Code: `github.com/mohamedhares/venturegraph` *(to be published)*
Report: `REPORT.md` · **Appendix:** `methodology_appendix.md`
Pre-registration: `preregistration.py` (SHA-256 locked 2026-05-10)
Paper draft (SSRN): `paper_abstract.md`

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